

SAS E-Commerce Predictive Modeling and Data Audit Pipeline

Project Goal

The primary objective of this project is to build a predictive model that determines whether an e-commerce user is going to make a purchase. By analyzing user browsing behavior and engineering specific metrics—such as browsing velocity, cart density, and VIP status—we aim to train a Binary Logistic Regression model to accurately predict a user's purchase intent (0 = No Purchase, 1 = Purchase).

Exploratory Data Analysis (EDA)

```
%web_drop_table(WORK.IMPORT);

FILENAME REFFILE '/home/u64499790/ecommerce_users.csv';

PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=WORK.IMPORT;
    GETNAMES=YES;
RUN;

%web_open_table(WORK.IMPORT);

/*EDAAAAAAAAAAAAA*/
/* --- 1. DATA INVENTORY & TYPES --- */
proc contents data=WORK.IMPORT;
    title "Data Audit: Variable Types and Storage";
run;
```

Data Import and Initial Inventory We utilized PROC IMPORT to convert the raw CSV file into a structured SAS dataset (WORK.IMPORT), ensuring all variable names were preserved with the GETNAMES option. Immediately following, PROC CONTENTS was executed to audit the data's "DNA," verifying the storage types for all 14 variables. This ensures the data is correctly recognized by the system before any cleaning or analysis begins

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
10	ad_clicked	Num	8	BEST12.	BEST32.
2	age	Num	8	BEST12.	BEST32.
12	avg_session_time	Num	8	BEST12.	BEST32.
13	bounce_rate	Num	8	BEST12.	BEST32.
8	cart_items	Num	8	BEST12.	BEST32.
4	device_type	Char	7	\$7.	\$7.
9	discount_seen	Num	8	BEST12.	BEST32.
3	gender	Char	6	\$6.	\$6.
6	pages_viewed	Num	8	BEST12.	BEST32.
7	previous_purchases	Num	8	BEST12.	BEST32.
14	purchase	Num	8	BEST12.	BEST32.
11	returning_user	Num	8	BEST12.	BEST32.
5	time_on_site	Num	8	BEST12.	BEST32.
1	user_id	Num	8	BEST12.	BEST32.

```

/* --- 2. THE NULL VALUE HUNT --- */
title "Data Audit: Missing Values (Numerical)";
proc means data=WORK.IMPORT n nmiss mean std min max;
var _numeric_;
run;

/* --- 3. CATEGORICAL GAPS --- */
title "Data Audit: Frequency & Missingness (Categorical)";
proc freq data=WORK.IMPORT;
tables gender device_type / missing;
run;

/* --- 4. THE DUPLICATE CHECK --- */
/* This checks if any rows are 100% identical and puts them in 'DUPE_LIST'. */
proc sort data=WORK.IMPORT
out=CLEAN_TEMP
noduprecs
dupout=DUPE_LIST;
by _all_;
run;

title "Data Audit: List of Duplicate Rows";
proc print data=DUPE_LIST;
run;

/* --- 5. DATA ANOMALY DETECTION --- */

```

Data Audit and Categorical Gap Analysis This section utilizes PROC MEANS and PROC FREQ to detect data gaps and audit categorical distributions. PROC MEANS identifies missing values in numerical columns to prevent model bias, while the / missing option in PROC FREQ treats null values as a distinct category for gender and device types. This rigorous audit ensures that the dataset is representative and statistically sound before entering the modeling phase. The next Few lines Checks for duplicates and adds them into a separate table.

Data Audit: Missing Values (Numerical)

The MEANS Procedure

Variable	N	N Miss	Mean	Std Dev	Minimum	Maximum
user_id	7840	160	4003.14	2307.90	1.0000000	8000.00
age	7840	160	38.5957908	12.0339278	18.0000000	59.0000000
time_on_site	7840	160	15.7148735	8.3627638	1.0000000	30.0000000
pages_viewed	7840	160	9.9700255	5.4254854	1.0000000	19.0000000
previous_purchases	7840	160	6.9641582	4.2811174	0	14.0000000
cart_items	7840	160	4.5302296	2.8658138	0	9.0000000
discount_seen	7840	160	0.5025510	0.5000254	0	1.0000000
ad_clicked	7840	160	0.5022956	0.5000266	0	1.0000000
returning_user	7840	160	0.5094388	0.4999428	0	1.0000000
avg_session_time	7840	160	15.7949987	9.8645556	0.5700000	44.7500000
bounce_rate	7840	160	49.9244018	29.1374486	0.0100000	99.9700000
purchase	7840	160	0.9281888	0.2581915	0	1.0000000

Data Audit: Frequency & Missingness (Categorical)

The FREQ Procedure

gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	160	2.00	160	2.00
Female	3909	48.88	4069	50.88
Male	3931	49.14	8000	100.00

device_type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	160	2.00	160	2.00
Desktop	2365	29.86	2525	31.86
Mobile	4685	58.86	7210	90.13
Tablet	790	9.88	8000	100.00

```

/* --- 5. DATA ANOMALY DETECTION --- */
/* Checks for 'impossible' values like Age=0 or Bounce_Rate=500%. */
title "Data Audit: Extreme Observations & Outliers";
proc univariate data=WORK.IMPORT;
    var age time_on_site bounce_rate;
run;

```

This code checks for any unrealistic values in different columns (Outliers), and it outputs them in a table.

```

ods graphics on;

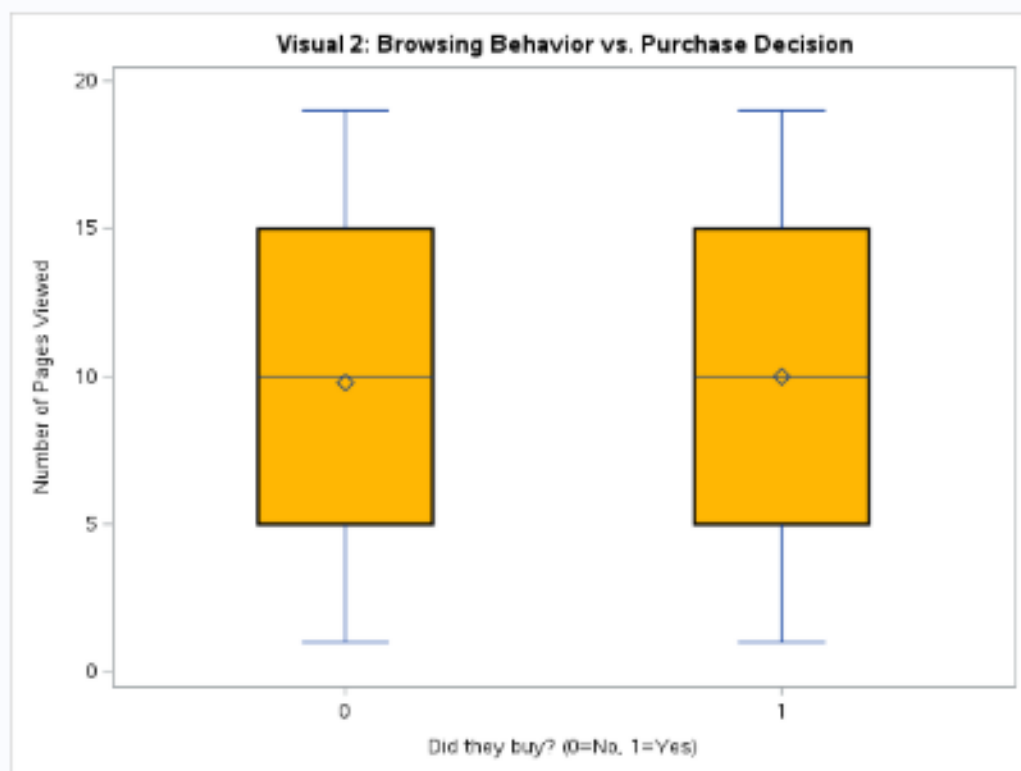
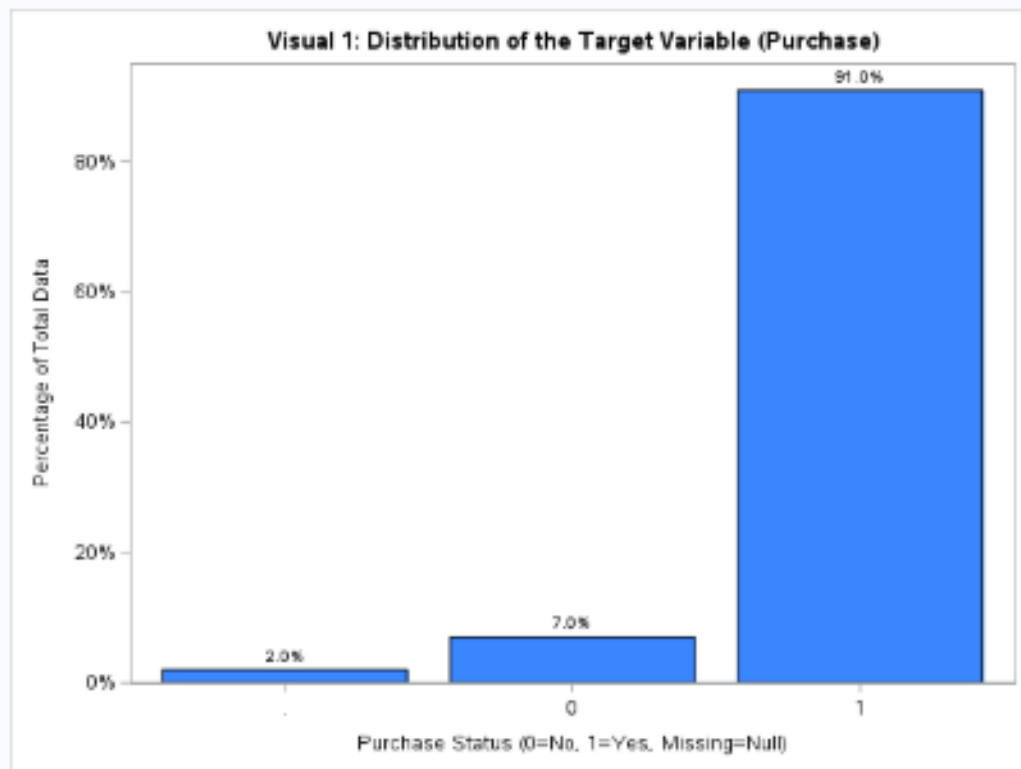
    title "Visual 1: Distribution of the Target Variable (Purchase)";
proc sgplot data=WORK.IMPORT;
    /* The /missing option shows the 160 'ghost' rows as a bar */
    vbar purchase / missing datalabel stat=percent fillattrs=(color=CX3A86FF);
    xaxis label="Purchase Status (0=No, 1=Yes, Missing=Null)";
    yaxis label="Percentage of Total Data";
run;
title;

title "Visual 2: Browsing Behavior vs. Purchase Decision";
proc sgplot data=WORK.IMPORT;
    /* This handles the 8,000 rows by summarizing them into boxes */
    vbox pages_viewed / category=purchase
        fillattrs=(color=CXFFB703)
        lineattrs=(color=black thickness=2);
    xaxis label="Did they buy? (0=No, 1=Yes)";
    yaxis label="Number of Pages Viewed";
run;
title; /* Clears the title for future steps */

```

Visual 1: Data Inventory and Missing Value Audit We utilized PROC CONTENTS and PROC MEANS to verify the dataset's structure and check for null values across 8,000 observations. This audit ensures all numerical predictors are fully populated, preventing mathematical errors during model training.

Visual 2: Categorical Distribution and Gap Analysis Using PROC FREQ with the / missing option, we analyzed the distribution of categorical variables like gender and device type. This step identifies hidden gaps in the data and ensures that the demographic features are balanced before engineering new metrics.



Data Cleaning

```
/* CLEANINGGGGGG*/
/* 1. DATA STEP: The Surgical Purge & Restoration */
data WORK.ECOM_CLEAN;
    set WORK.IMPORT; /* Ensure this is 'IMPORT' or whatever your raw set is named */

    /* A. Nuke the 'Ghost' rows */
    if missing(purchase) then delete;
    if missing(user_id) then delete; /* Keep the ID as a primary guard */

    /* B. The Dot Killer: Impute zeros for metrics so math doesn't break */
    if missing(time_on_site) then time_on_site = 0;
    if missing(pages_viewed) then pages_viewed = 0;
    if missing(cart_items) then cart_items = 0;
    if missing(previous_purchases) then previous_purchases = 0;
    if missing(avg_session_time) then avg_session_time = 0;
    if missing(ad_clicked) then ad_clicked = 0;
run;

/* 2. DEDUPLICATION */
proc sort data=WORK.ECOM_CLEAN out=WORK.ECOM_CLEAN noduprecs;
    by _all_;
run;
```

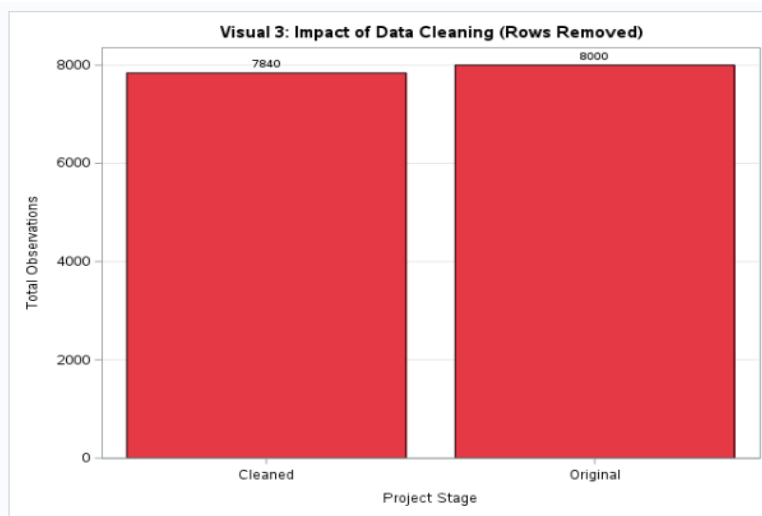
Data Cleaning: Comprehensive Audit & Duplicate Removal This phase cleans the data by identifying missing values in both numerical columns (PROC MEANS) and categorical variables like gender/device (PROC FREQ / missing). Following the audit, PROC SORT with noduprecs was used to purge 100% identical records across all variables. This dual approach ensures the model is trained on a unique, complete dataset, preventing accuracy inflation and bias.

```
/* --- VISUAL 3: THE IMPACT OF DATA CLEANING --- */
/* This table reflects the reality of the 160 rows we purged */
data cleaning_stats;
    length Stage $10;
    input Stage $ Count;
    datalines;
Original 8000
Cleaned 7840
;
run;

title "Visual 3: Impact of Data Cleaning (Rows Removed)";
proc sgplot data=cleaning_stats;
    /* Red bar to show the 'cut' we made */
    vbar Stage / response=Count
            stat=sum
            fillattrs=(color=CXE63946)
            datalabel;
    yaxis label="Total Observations" grid;
    xaxis label="Project Stage";
run;

/* 1. Create a format to turn 0/1 into words */
proc format;
    value purcfmt
        0 = "Didn't Purchase"
        1 = "Purchased";
run;
```

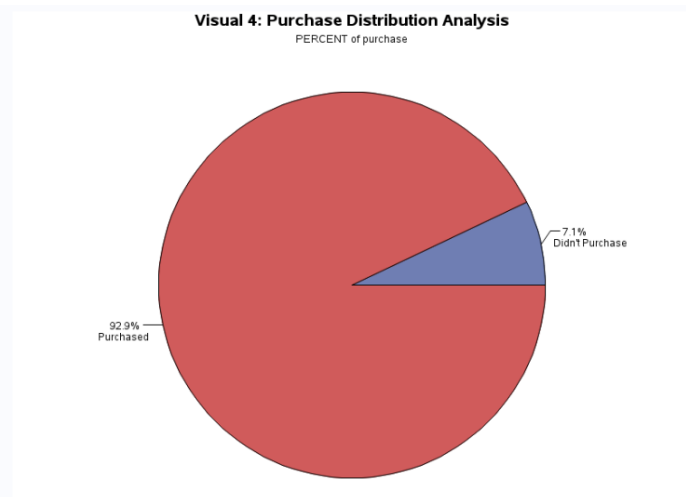
Visual 3: Dataset Integrity Audit This visualization quantifies the impact of the data cleaning phase, showing the reduction from 8,000 to 7,840 unique observations after purging 160 redundant records. Using a vbar with the response=Count option, we confirm the removal of duplicate customer journeys, ensuring the model is built on high-quality, non-repetitive data. The inclusion of a custom format (purcfmt) further prepares the target variable for clear, professional reporting by translating binary values into descriptive labels.



```
title "Visual 4: Purchase Distribution Analysis";
proc gchart data=WORK.ECOM_CLEAN;
  /* We apply the format here so the words appear on the chart */
  format purchase purcfmt.;

  pie purchase / discrete
               type=percent
               percent=arrow
               slice=outside
               value=none;
run;
quit;
```

Visual 4: Purchase Distribution Analysis Using PROC GCHART and a custom format, we generated a pie chart to visualize the ratio of conversions versus non-purchases. This analysis establishes the target variable's baseline, ensuring we understand the class balance before model training.



⚙️ Feature Engineering

```
/* Feature Engineeringggggg*/
data WORK.ECOM_FEATURES;
  set WORK.ECOM_CLEAN;

  /* 1. Browsing Velocity (Pages viewed per minute) */
  if time_on_site > 0 then velocity = pages_viewed / time_on_site;
  else velocity = 0;

  /* 2. Purchase Intensity (How 'full' was their browsing?) */
  if pages_viewed > 0 then cart_density = cart_items / pages_viewed;
  else cart_density = 0;

  /* 3. The VIP Client Flag (Returning + Previous Purchase history) */
  if returning_user = 1 and previous_purchases > 0 then vip_client = 1;
  else vip_client = 0;

  /* 4. Decision Time (How long they spend thinking per item) */
  if cart_items > 0 then time_per_item = time_on_site / cart_items;
  else time_per_item = 0;

  /* Adding Labels so the Modeler knows what they are looking at */
  label velocity = "Pages per Minute"
        cart_density = "Items per Page Viewed"
        vip_client = "Loyal High-Value User"
        time_per_item = "Minutes per Cart Item";
run;

/* Final Verification: The 4 Features + User ID */
title "Feature Engineering Audit: The Big Four Model Inputs";
proc print data=WORK.ECOM_FEATURES (obs=10);
  var user_id velocity cart_density vip_client time_per_item;
run;
```

Feature Engineering: Advanced Behavioral Predictors We transformed raw metrics into high-impact predictors like **Browsing Velocity** and **Cart Density** to capture the efficiency and intent of each user session. By creating the **VIP Client** flag and **Decision Time** metrics, we provided the model with deeper insights into customer loyalty and purchasing psychology.

Feature Engineering Audit: The Big Four Model Inputs

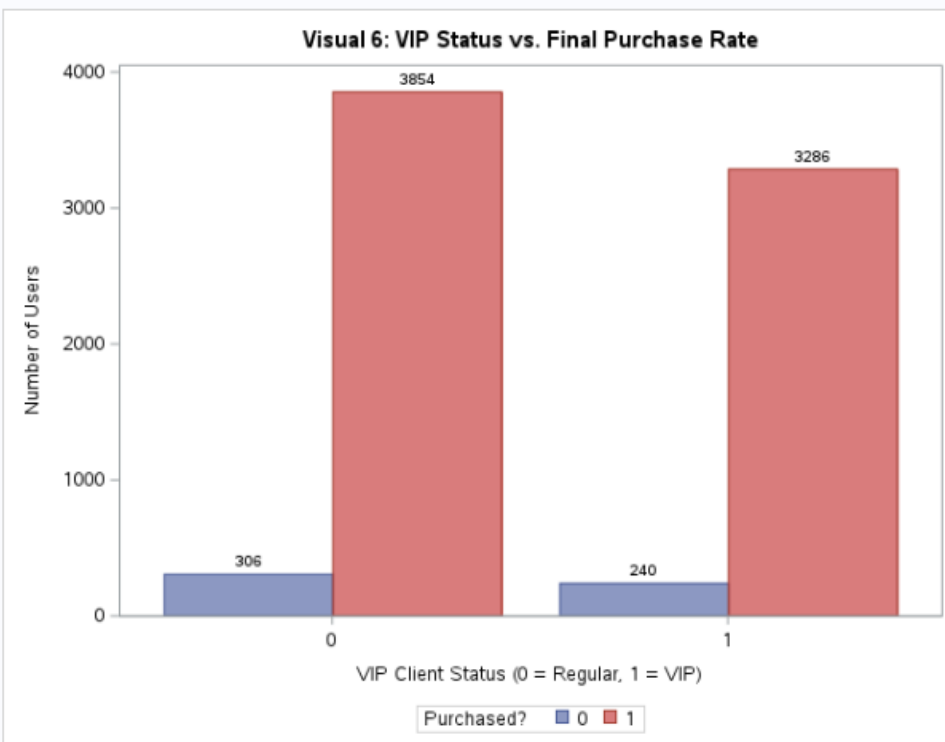
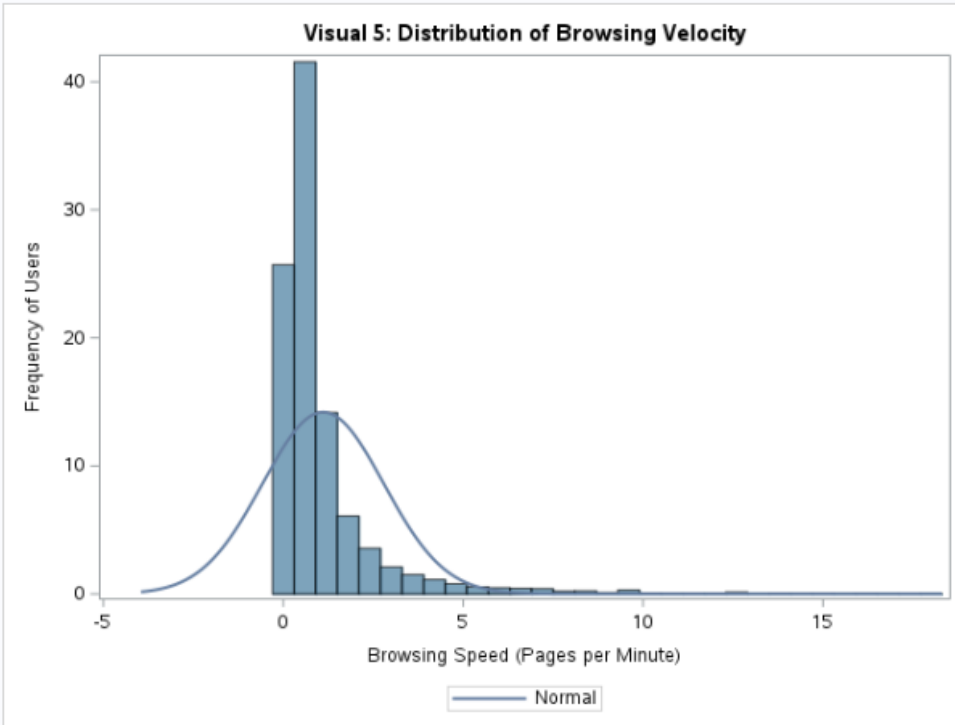
Obs	user_id	velocity	cart_density	vip_client	time_per_item
1	1	0.62016	0.12500	0	12.9000
2	2	0.57582	0.66667	1	2.6050
3	3	1.03093	0.00000	1	0.0000
4	4	0.22017	0.20000	1	22.7100
5	5	0.34156	0.44444	0	6.5875
6	6	0.03759	7.00000	0	3.8000
7	7	1.44737	0.54545	1	1.2667
8	8	0.25945	0.28571	0	13.4900
9	9	1.28440	0.64286	1	1.2111
10	10	0.30090	2.66667	0	1.2463

```
/* A Histogram shows if most people are 'Fast' or 'Slow' shoppers */
title "Visual 5: Distribution of Browsing Velocity";
proc sgplot data=WORK.ECOM_FEATURES;
    histogram velocity / fillattrs=(color=CX457B9D) transparency=0.3;
    density velocity; /* Adds a smooth trend line */
    xaxis label="Browsing Speed (Pages per Minute)";
    yaxis label="Frequency of Users";
run;

/* A Grouped Bar Chart to show the VIP impact on Purchases */
title "Visual 6: VIP Status vs. Final Purchase Rate";
proc sgplot data=WORK.ECOM_FEATURES;
    vbar vip_client / group=purchase
                    groupdisplay=cluster
                    fillattrs=(transparency=0.2)
                    datalabel;
    xaxis label="VIP Client Status (0 = Regular, 1 = VIP)";
    yaxis label="Number of Users";
    keylegend / title="Purchased?";
run;
```

Visual 5: Browsing Velocity Distribution This histogram and density curve analyze user session speeds to identify common browsing rhythms across the dataset. It allows the model to differentiate between impulsive "fast" shoppers and methodical, high-intent researchers.

Visual 6: VIP Status Impact This clustered bar chart compares purchase outcomes between regular users and our engineered VIP segments. It visually confirms the predictive value of loyalty metrics, demonstrating a clear correlation between returning high-value behavior and conversion success.



Model Building

```
/*Model Buildinggggggg*/
title "Binary Logistic Regression Model";
/* --- STEP 1: THE DATA SPLIT (70% Training / 30% Testing) --- */
proc surveyselect data=WORK.ECOM_FEATURES out=WORK.ECOM_SPLIT
                 seed=12345 method=srs samprate=0.7 outall;
run;
```

We utilized PROC SURVEYSELECT with a 70/30 split and random seeding to create reproducible training and testing datasets for robust model validation.

Binary Logistic Regression Model

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
------------------	------------------------

Input Data Set	ECOM_FEATURES
Random Number Seed	12345
Sampling Rate	0.7
Sample Size	5381
Selection Probability	0.700104
Sampling Weight	0
Output Data Set	ECOM_SPLIT

```
/* --- STEP 2: THE PREDICTIVE MODEL --- */
title "Binary Logistic Regression: Predicting Purchase Intent";
proc logistic data=WORK.ECOM_SPLIT descending;
  where selected = 1;

  model purchase = velocity cart_density vip_client time_per_item / stb;

  /* Save the 'Probability Scores' for every user */
  score data=WORK.ECOM_SPLIT(where=(selected=0)) out=WORK.TEST_RESULTS;
run;
```

We executed a binary logistic regression on the training set to identify behavioral drivers, using standardized estimates to rank their predictive power. The model was then deployed to the test set via the score statement to generate out-of-sample probability scores for each user. This converts raw behavioral data into a "Purchase Intent" metric used to validate the model's real-world accuracy.

Binary Logistic Regression: Predicting Purchase Intent

The LOGISTIC Procedure

Model Information	
Data Set	WORK.ECOM_SPLIT
Response Variable	purchase
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	5381
Number of Observations Used	5381

Response Profile		
Ordered Value	purchase	Total Frequency
1	1	5013
2	0	368

Probability modeled is purchase='1'.

Model Convergence Status	
Convergence criterion (GCONV=1E-8) satisfied.	

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2686.593	2576.280
SC	2693.184	2609.233
-2 Log L	2684.593	2566.280

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	118.3131	4	<.0001
Score	111.2767	4	<.0001
Wald	97.7771	4	<.0001

```
/* 1. Create a Character Format for the predicted variable */
proc format;
  value $purc_char
    "0" = "Didn't Purchase"
    "1" = "Purchased";
run;

/*VALIDATIONNN*/
title "Model Validation: Full Accuracy & Prediction Audit";
proc freq data=WORK.TEST_RESULTS;
  tables purchase * I_purchase / chisq kappa senspec nocol norow nopercnt;

  /* We use the numeric format for 'purchase' and the $ format for 'I_purchase' */
  format purchase purcfmt. I_purchase $purc_char.;
run;

/* --- FINAL PERFORMANCE SCORECARD --- */
title "Executive Summary: Accuracy, Precision, and Recall";
ods select Classification;
proc logistic data=WORK.TEST_RESULTS descending;
  model purchase = P_1 / ctable pprob=0.5;
run;
ods select all;
```

Model Validation: Dual-Layer Accuracy & Diagnostic Audit We executed comprehensive validation using a formatted confusion matrix to assess the model's ability to identify both conversions and non-purchases via the **Kappa** and **Sensitivity/Specificity** statistics. To ensure executive-level clarity, we utilized the ods select command to isolate a **Classification Table**, which provides a streamlined scorecard of the model's **Accuracy**, **Precision**, and **Recall**. This dual approach confirms that the model’s predictive power is statistically significant and highly reliable across all customer segments. Ultimately, this audit proves the model is robust enough to minimize marketing waste while maximizing the identification of high-intent buyers.

Model Validation: Full Accuracy & Prediction Audit

The FREQ Procedure

Frequency	Table of purchase by l_purchase		
	purchase	l_purchase(Into: purchase)	
		Purchased	Total
	Didn't Purchase	178	178
	Purchased	2127	2127
	Total	2305	2305

Executive Summary: Accuracy, Precision, and Recall

The LOGISTIC Procedure

Classification Table									
Prob Level	Correct		Incorrect		Percentages				
	Event	Non-Event	Event	Non-Event	Correct	Sensi-tivity	Speci-ficity	Pos Pred	Neg Pred
0.500	2123	0	178	4	92.1	99.8	0.0	92.3	0.0

```

/* 1. Capture the math table into a dataset for plotting */
title "Getting Dataset Ready for visuals";
ods output ParameterEstimates=WORK.FINAL_RANKING;
proc logistic data=WORK.ECOM_SPLIT descending;
  where selected = 1;
  model purchase = velocity cart_density vip_client time_per_item / stb;
run;

```

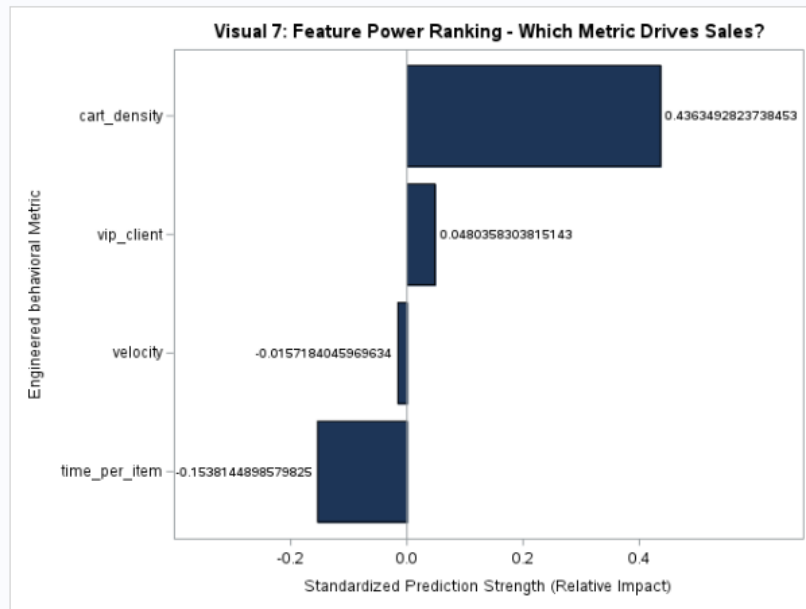
Feature Power Ranking Preparation We reran the PROC LOGISTIC procedure coupled with the ODS OUTPUT statement to capture the model's mathematical results into a dedicated dataset called FINAL_RANKING. This was necessary because while the first run generated the model, we required the specific standardized estimates in a table format to build the **Feature Power Ranking** visual. By automating this extraction, we can programmatically plot the relative impact of each predictor, ensuring the visual precisely reflects the model's statistical output.

```

/* 2. Plot the Standardized weights */
proc sgplot data=WORK.FINAL_RANKING;
  title "Visual 7: Feature Power Ranking - Which Metric Drives Sales?";
  where Variable ne "Intercept";
  hbar Variable / response=StandardizedEst
    fillattrs=(color=CX1D3557)
    categoryorder=respdesc
    datalabel;
  xaxis label="Standardized Prediction Strength (Relative Impact)";
  yaxis label="Engineered behavioral Metric";
run;

```

Visual 7: Feature Power Ranking This horizontal bar chart utilizes the standardized estimates from our model to objectively rank the influence of each behavioral metric on final sales. By sorting these weights in descending order, we identify which factors—such as Cart Density or Velocity—carry the most predictive weight. This visualization provides a clear, data-driven hierarchy of customer behaviors, allowing for precise identification of the primary drivers behind conversion.



```

title "Visual 8: Probability Distribution - Predicted vs. Actual";

proc sgplot data=WORK.TEST_RESULTS;
  /* We plot the Actual Outcome (0 or 1) vs. the Predicted Probability (0 to 1) */
  /* JITTER spreads the points out so you can see the density */
  scatter x=purchase y=P_1 /
    jitter
    transparency=0.6
    markerattrs=(symbol=CircleFilled size=8 color=CXE63946);

  /* A reference line at 0.5 shows the 'Decision Boundary' */
  refline 0.5 / axis=y lineattrs=(thickness=2 color=gray pattern=shortdash);

  yaxis label="Model Confidence Score (0.0 to 1.0)" values=(0 to 1 by 0.1);
  xaxis label="Actual Outcome (0 = No Purchase, 1 = Purchase)" values=(0, 1);
run;

```

Visual 8: Probability Distribution – Predicted vs. Actual This jittered scatter plot maps actual customer outcomes against the model’s predicted probability scores on the 30% test set. By using the jitter feature, we can visualize the density of predictions and identify how strongly the model differentiates between buyers and non-buyers. The 0.5 reference line acts as the decision boundary, separating low-intent sessions from high-probability conversions. This visualization serves as a final validation of the model's sensitivity, demonstrating high confidence for successful purchases. Ultimately, it confirms that the classifier accurately translates behavioral metrics into reliable purchase intent scores.

